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# Long-Term Mortality, Cognition, and Quality of Life After Chronic Subdural Hematoma Surgery

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Editorial

Supplemental content

**IMPORTANCE** Chronic subdural hematoma (cSDH) is among the most common neurosurgical disorders in older adults. Although short-term outcomes after surgery are favorable, long-term survival and health-related quality of life (HRQoL) remain poorly characterized.

**OBJECTIVE** To evaluate long-term survival, excess mortality, and HRQoL 10 years after surgical treatment of cSDH.

**DESIGN, SETTING, AND PARTICIPANTS** This population-matched cohort study was conducted at a single tertiary referral center in Switzerland, with mortality follow-up through December 31, 2023 (mean [SD] follow-up, 9.55 [1.24] years), and cross-sectional HRQoL assessment through December 31, 2024 (mean [SD] follow-up, 10.05 [1.16] years). Analyses were conducted from October to December 2025. Adults surgically treated for cSDH between June 2012 and August 2016 were included, matched with the Swiss general population by age, sex, and birth month for mortality analysis. Among survivors, those completing HRQoL assessment were compared with age- and sex-weighted European reference values.

**EXPOSURE** Surgically treated cSDH.

**MAIN OUTCOMES AND MEASURES** The primary outcome was all-cause mortality, estimated using Kaplan-Meier analysis, with excess mortality expressed as absolute survival differences and standardized mortality ratios (SMRs). Secondary outcomes were the following HRQoL domains: cognitive functioning (CF), physical functioning (PF), role functioning (RF), emotional functioning (EF), social functioning (SF), and global QoL, compared using 2-sided *z* tests.

**RESULTS** A total of 359 adults surgically treated for cSDH were included; among survivors, 147 completed HRQoL assessment and were compared with age- and sex-weighted European reference values. Among 359 patients (mean [SD] age, 73.4 [11.0] years; 117 female patients [32.6%]), overall survival was significantly lower than matched controls (hazard ratio [cohort vs control], 2.02; 95% CI, 1.73-2.37; log-rank $P < .001$). One-year survival in the cSDH cohort was 92.8% (95% CI, 90.1%-95.5%) vs 98.8% (95% CI, 98.7%-98.8%) in controls, representing an excess mortality of 6.0 percentage points (SMR, 3.22; 95% CI, 2.10-4.72); 5-year survival was 76.6% (95% CI, 72.3%-81.1%) vs 88.2% (95% CI, 88.2%-88.3%), representing an excess of 11.6 percentage points (SMR, 1.19; 95% CI, 0.95-1.47); and 10-year survival was 55.5% (95% CI, 50.3%-61.3%) vs 73.5% (95% CI, 73.4%-73.6%), representing an excess of 18.0 percentage points (SMR, 1.12; 95% CI, 0.94-1.31). Men reported significantly lower mean (SD) PF scores (75.9 [26.8] vs control mean score, 83.22; $P < .001$), RF scores (74.9 [32.0] vs 84.87; $P < .001$), CF scores (77.6 [22.6] vs 87.38; $P < .001$), and SF scores (84.3 [24.0] vs 90.00; $P = .02$) than controls, and women reported lower mean (SD) RF (69.0 [30.9] vs 80.91; $P = .02$) and CF scores (70.2 [24.8] vs 86.50; $P < .001$). EF and global QoL did not differ significantly from European reference values.

**CONCLUSIONS AND RELEVANCE** In this population-matched cohort study, patients surgically treated for cSDH experienced sustained excess mortality and clinically relevant HRQoL deficits 10 years after surgery. These findings call for structured postoperative and rehabilitative care beyond the acute phase.

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Chronic subdural hematoma (cSDH) is among the most common neurosurgical conditions in older individuals, with incidence rates more than doubling over the past 2 decades and projected to rise further with population aging and widespread anticoagulation use.[1,2] Although surgical evacuation is typically effective and short-term outcomes are favorable, cSDH is increasingly recognized not as a transient surgical problem but as a manifestation of broader frailty and cerebral vulnerability.[3-6]

Several studies have reported elevated mortality after cSDH surgery, with 1-year mortality rates reaching up to 24%.[3,4,7-9] However, most available data are limited to short-term follow-up, and robust longitudinal evidence beyond 5 years remains scarce. Therefore, the long-term survival trajectory and whether excess mortality persists remains unknown.

Health-related quality of life (HRQoL) encompasses subjective higher-order dimensions of functioning that are not captured by conventional disability scales. Although several studies have assessed HRQoL following cSDH, follow-up is typically limited to the first 3 to 6 months, offering only a view on early postoperative recovery.[10,11] Long-term outcomes, particularly among patients surviving many years after treatment, remain largely unexplored.[12] In a population at potentially increased risk for mortality, evaluating the self-perceived cognition, function, and health status of long-term survivors is essential not only to identify persistent deficits that may escape standard clinical assessments, but also to determine whether this group represents a resilient subset with preserved quality of life or individuals living with subtle but enduring impairments.

To address these gaps, a population-matched cohort study of patients with surgically treated cSDH originally enrolled in a prospective trial was conducted, with a mean follow-up of 10 years. Our objective was to quantify long-term survival and HRQoL compared with the general population and to provide contemporary benchmark data on the enduring impact of cSDH.

### Key Points

**Question** What are the long-term survival and health-related quality of life outcomes in patients surgically treated for chronic subdural hematoma (cSDH)?

**Findings** In this 10-year follow-up cohort study of 359 patients, survival was 18 percentage points lower than in an age- and sex-matched general population (55.5% vs 73.5%), and survivors reported persistent cognitive and functional impairments despite overall preserved global quality of life.

**Meaning** Patients with surgically treated cSDH experience sustained excess mortality and long-term self-reported cognitive and functional impairments, underscoring the need for comprehensive postoperative and rehabilitative care beyond the perioperative episode.

## Methods

### Ethical Approval

This study was approved by the local ethics committee (2023-01568) and conducted in accordance with the Declaration of Helsinki.[13] Data were deidentified prior to analysis, and written or verbal informed consent was obtained for the HRQoL assessment.

### Study Design and Participants

This study was a population-matched cohort study with cross-sectional follow-up of long-term survivors and is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.[14] Patients with cSDH who underwent surgical evacuation between June 5, 2012, and August 1, 2016, were identified from a previous randomized clinical trial.[15] Of the 361 patients originally enrolled, 359 patients with complete data were included in the present analysis. All underwent surgical evacuation of cSDH according to standard institutional protocols described earlier.[15,16]

### Population Matching, Survival, and Excess Mortality

All-cause mortality in patients with cSDH was ascertained through the hospital registry and verified against the Swiss national death registry, which is updated monthly and linked through unique social security numbers, ensuring complete data accuracy.

For comparative analysis, survival was derived from a demographically matched reference cohort based on nationwide data provided by the Swiss Federal Statistical Office (FSO).[17] This dataset encompasses the entire resident population, containing anonymized records of all Swiss residents from January 1, 2011, to December 31, 2023, including dates of death. For each patient, control individuals were matched by age, sex, and birth month and were required to be alive at the date of the patient's inclusion in the initial study. Survival analysis was censored at December 31, 2023, the latest date of complete mortality data. Mortality data from matched controls were used to generate population-expected survival curves. For patients with cSDH, causes of death were identified through national registry linkage provided by the Swiss FSO[18] and coded according to the *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10)* chapter and reported descriptively (eTable 1 in Supplement 1).

### HRQoL

HRQoL was assessed cross-sectionally through December 31, 2024, using the European Organization for Research and Treatment of Cancer Quality of Life Questionnaire (EORTC QLQ-C30, version 3.0),[19] a validated 30-item instrument. Only patients who were alive at the time of follow-up were eligible and contacted for participation. Patients were first contacted by letter and subsequently by telephone after providing consent.

The following 6 functional domains were calculated: physical functioning (PF), role functioning (RF), emotional functioning (EF), cognitive functioning (CF), social functioning (SF), and global QoL. Each domain comprises 2 to 5 items rated on 4-point Likert scales and converted to domain scores from 0 (worst) to 100 (best).[20] CF is assessed through items on memory and concentration difficulties, whereas PF, RF, and SF evaluate limitations in mobility, daily activities, and social participation, respectively.

Normative reference values for comparison were obtained from the European population dataset published by Nolte and colleagues.[21] Because EORTC QLQ-C30 scores vary substantially by age and sex, expected population means were generated through age- and sex-weighted standardization. Each sex-specific age group of our cohort was matched to the corresponding normative stratum, and weighted population means and standard deviations were calculated using the cohort's distribution (eTable 2 in Supplement 1). To ensure robustness, the weighting procedure was tested using an alternative stratification rule with exclusion of very small strata. Full details are provided in the eAppendix in Supplement 1.

### Statistical Analysis

Kaplan-Meier survival estimates were calculated with corresponding 95% confidence intervals for both the cSDH cohort and the matched Swiss population cohort, and group differences were evaluated using the log-rank test. Excess mortality at 1, 5, and 10 years was calculated as the absolute difference in survival probability between groups. In addition, standardized mortality ratios (SMRs) were calculated at the same time points by comparing observed deaths in the cSDH cohort with expected deaths derived from individually matched population controls using the Byar method implemented in the *ems* R package. Comparisons of EORTC QLQ-C30 scores between the cohort and age- and sex-adjusted reference data were performed using 2-sided 1-sample $z$ tests for each domain. $P$ values were adjusted for multiple comparisons using the Holm method. Statistical significance was defined as a 2-sided $P < .05$. All analyses were performed using R version 4.5.0 (R Foundation) and RStudio version 2024.12.1+563 (Kousa Dogwood build), with SQL queries executed through the DuckDB engine.

## Results

### Baseline Characteristics

A total of 359 patients were included in the analysis (117 female patients [32.6%]; mean [SD] age, 73.4 [11.0] years). The median (IQR) Glasgow Coma Scale score at admission was 15 (15-15), and the median (IQR) National Institutes of Health Stroke Scale score was 2 (0-3). Most patients (330 [91.9%]) lived independently at admission. Arterial hypertension (197 [55.0%]), coronary artery disease (98 [27.4%]), and cardiac arrhythmia (91 [25.4%]) were the most common comorbidities. Detailed baseline characteristics are summarized in **Table 1**.

### Survival and Excess Mortality

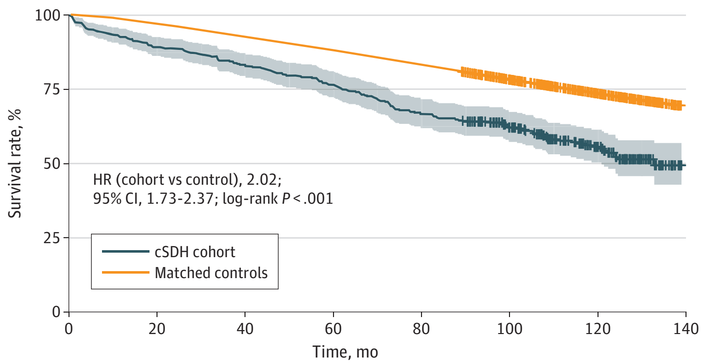
Long-term survival data until December 31, 2023, were available for all 359 patients. During a mean (SD) follow-up of 9.55 (1.24) years, 153 patients (42.6%) died at a median (IQR) age of 85.2 (79.1-90.1) years. Population matching yielded 919 625 control individuals, with a median (range) of 2415 (58-6276) controls per patient (eFigure in Supplement 1). Survival analysis showed sustained excess long-term mortality among patients with cSDH compared with matched controls (hazard ratio [cohort vs control], 2.02; 95% CI, 1.73-2.37; log-rank $P < .001$)

Table 1. Baseline Characteristics of the Chronic Subdural Hematoma Cohort (N = 359) at Time of Inclusion

| Characteristic | No. (%) |
|---|---|
| Age, mean (SD), y | 73.4 (11.0) |
| Sex | |
| Men | 242 (67.4) |
| Women | 117 (32.6) |
| GCS at admission, median (IQR) | 15 (15-15) |
| NIHSS at admission, median (IQR) | 2 (0-3) |
| Living situation at admission | |
| Independent (with or without partner) | 330 (91.9) |
| Independent with home care | 14 (3.9) |
| Assisted living | 9 (2.5) |
| Nursing home | 3 (0.8) |
| Medical facility | 3 (0.8) |
| Comorbidities | |
| None | 77 (21.5) |
| Hypertension | 197 (55.0) |
| Alcohol use disorder | 13 (3.6) |
| COPD | 12 (3.4) |
| Cerebrovascular disease | 26 (7.3) |
| Dementia | 10 (2.8) |
| Diabetes | 57 (15.9) |
| Epilepsy | 11 (3.1) |
| Cardiac arrhythmias | 91 (25.4) |
| Coronary artery disease | 98 (27.4) |
| Nicotine use disorder | 18 (5.0) |
| Malignancy (history) | 41 (11.5) |
| Deep vein thrombosis | 22 (6.1) |
| Antithrombotic therapy | |
| None | 169 (47.5) |
| Aspirin | 92 (25.8) |
| Clopidogrel | 15 (4.2) |
| Oral anticoagulants | 96 (27.0) |
| Low-molecular-weight heparin | 3 (0.8) |
| Admission symptoms | |
| Gait disturbance | 175 (49.9) |
| Headache | 221 (63.0) |
| Hemiparesis | 106 (30.2) |
| Speech disturbance | 80 (22.8) |
| Confusion | 67 (19.1) |
| Modified Rankin Scale score at admission | |
| 0-1 | 187 (52.1) |
| 2-3 | 147 (40.9) |
| 4-5 | 25 (7.0) |
| Mobility at admission | |
| Independent | 272 (75.8) |
| Cane or crutches | 24 (6.7) |
| Walker or assistance | 27 (7.5) |
| Wheelchair | 5 (1.4) |
| Bedridden | 31 (8.6) |
| Traumatic incidents in recent months | |
| Fall with head trauma | 197 (55.6) |
| Fall without head trauma | 48 (13.6) |
| Fall with loss of consciousness | 26 (7.3) |
| Head trauma (without fall) | 17 (4.8) |
| No fall | 36 (10.2) |
| Unknown | 36 (10.2) |

Abbreviations: COPD, chronic obstructive pulmonary disease; GCS, Glasgow Coma Scale; NIHSS, National Institutes of Health Stroke Scale.

Figure 1. Kaplan-Meier Curve: Chronic Subdural Hematoma (cSDH) Cohort vs Matched General Population




| No. at risk | | | | | | | |
|---|---|---|---|---|---|---|---|
| cSDH cohort | 359 | 321 | 298 | 275 | 239 | 179 | 91 |
| Matched controls | 919625 | 893411 | 853539 | 811431 | 765770 | 578614 | 293208 |

HR indicates hazard ratio.

Table 2. Health-Related Quality of Life of Women (n = 42) and Men (n = 105) at 10-Year Follow-Up Compared to the European Reference Cohort

| Scale | cSDH cohort, mean (SD) | European reference, mean | *z* Value | *P* value |
|---|---|---|---|---|
| Women | | | | |
| PF | 74.6 (24.5) | 78.90 | −1.39 | .83 |
| RF | 69.0 (30.9) | 80.91 | −2.92 | .02 |
| EF | 79.2 (20.4) | 79.38 | −0.07 | >.99 |
| CF | 70.2 (24.8) | 86.50 | −6.06 | <.001 |
| SF | 83.7 (19.9) | 88.71 | −1.55 | .73 |
| gQoL | 69.0 (17.4) | 64.64 | 1.35 | .83 |
| Men | | | | |
| PF | 75.9 (26.8) | 83.22 | −3.85 | <.001 |
| RF | 74.9 (32.0) | 84.87 | −4.44 | <.001 |
| EF | 84.6 (20.7) | 83.63 | 0.57 | >.99 |
| CF | 77.6 (22.6) | 87.38 | −6.09 | <.001 |
| SF | 84.3 (24.0) | 90.00 | −2.97 | .02 |
| gQoL | 69.0 (20.1) | 69.06 | −0.01 | >.99 |

Abbreviations: CF, cognitive functioning; cSDH, chronic subdural hematoma; EF, emotional functioning; gQoL, global quality of life; PF, physical functioning; RF, role functioning; SF, social functioning.

(**Figure 1**). At 1 year, survival was 92.8% (95% CI, 90.1%-95.5%) in the cSDH cohort vs 98.8% (95% CI, 98.7%-98.8%) in controls, corresponding to an excess mortality of 6.0 percentage points and an SMR of 3.22 (95% CI, 2.10-4.72). At 5 years, survival was 76.6% (95% CI, 72.3%-81.1%) vs 88.2% (95% CI, 88.2%-88.3%), representing an excess mortality of 11.6 percentage points (SMR, 1.19; 95% CI, 0.95-1.47). At 10 years, survival was 55.5% (95% CI, 50.3%-61.3%) vs 73.5% (95% CI, 73.4%-73.6%), yielding an excess mortality of 18.0 percentage points (SMR, 1.12; 95% CI, 0.94-1.31). Causes of death among patients with cSDH were heterogeneous, most commonly cardiovascular, neurological, and malignant, and are summarized descriptively in eTable 1 in Supplement 1.

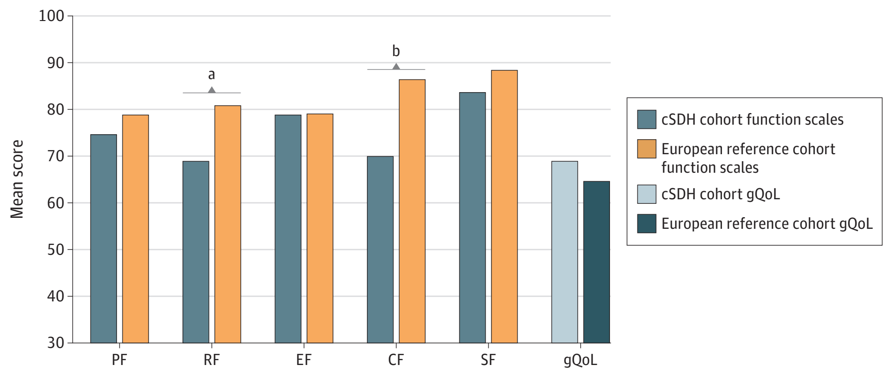
## HRQoL

The mean (SD) follow-up time from admission to the date of HRQoL assessment, by which all questionnaires had been completed, was 10.05 (1.16) years. Of patients alive at follow-up (n = 193), 147 (76.2%) completed the questionnaire. In men, mean (SD) PF, RF, CF, and SF scores were significantly lower than population norms (PF: 75.9 [26.8] vs 83.22; *P* < .001; RF: 74.9 [32.0] vs 84.87; *P* < .001; CF: 77.6 [22.6] vs 87.38; *P* < .001; and SF: 84.3 [24.0] vs 90.00; *P* = .02), while EF and global QoL were comparable. In women, RF and CF were significantly lower (RF: 69.0 [30.9] vs 80.91; *P* = .02; CF: 70.2 [24.8] vs 86.50; *P* < .001), whereas PF and SF were lower but did not differ significantly from population norms. Overall, survivors demonstrated preserved emotional well-being but measurable deficits in cognitive and functional domains. Results are summarized in **Table 2** dichotomized by sex and visualized in **Figure 2** and **Figure 3**. The alternative stratification rule yielded the same results (eTable 3 in Supplement 1).

## Discussion

In this long-term follow-up cohort study of surgically treated patients with cSDH, sustained excess mortality and persistent deficits of HRQoL a decade after treatment were found. At 10 years, only 55.5% of patients were alive compared with

Figure 2. Bar Graph Showing Health-Related Quality of Life of Women (n = 42) at 10-Year Follow-Up Compared to the European Reference Cohort

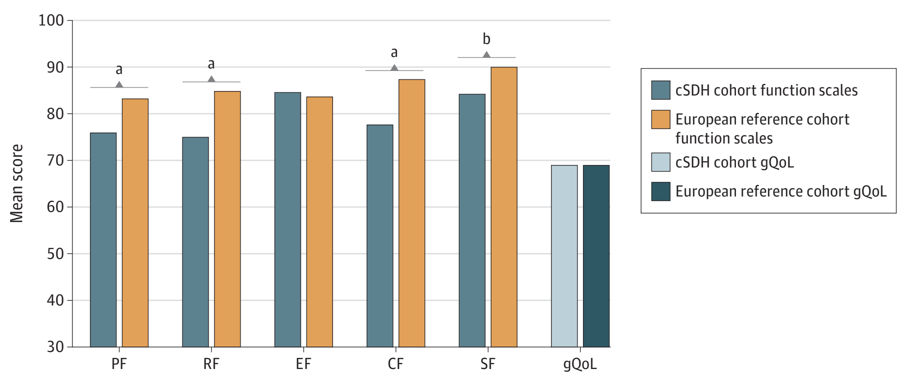



CF indicates cognitive functioning; EF, emotional functioning; gQoL, global quality of life; PF, physical functioning; RF, role functioning; SF, social functioning.

[a]$P = .02$.

[b]$P < .001$.

Figure 3. Health-Related Quality of Life of Men (n = 105) at 10-Year Follow-Up Compared to the European Reference Cohort




CF indicates cognitive functioning; EF, emotional functioning; gQoL, global quality of life; PF, physical functioning; RF, role functioning; SF, social functioning.

[a]$P < .001$.

[b]$P = .02$.

73.5% of matched individuals from the general population, representing an excess mortality of 18 percentage points. While this absolute survival gap increased with time, SMRs indicate that the relative excess risk is greatest during the first year after surgery and attenuates thereafter, reflecting complementary perspectives on cumulative vs time-dependent mortality risk. Notably, 1-year survival in our cohort was 92.8%, higher than previously reported rates ranging from 75% to 87% in comparable studies,[3-5,7-9] a difference unlikely to be explained by age given the similar mean age at enrollment relative to prior cohorts. This advantage may be attributable to variations in study design, as other studies often included conservatively managed, mixed patient populations and were conducted retrospectively, thereby limiting comparability. In fact, the relatively low early mortality and the baseline characteristics of this study's patients, who were predominantly living independently, indicate that the cohort was not systematically disadvantaged by frailty or socioeconomic factors. Even if the underlying trial cohort included patients with more favorable baseline health, the observed excess mortality would just represent a conservative estimate and may therefore even underestimate the true long-term mortality. This strengthens the validity of our findings and underscores that patients with cSDH remain at elevated risk of death well beyond the perioperative period.

The aim of this study was to establish reliable estimates of long-term survival in surgically treated patients with cSDH. Health trajectories, frailty, and comorbidities evolve substantially over time, and modeling their causal effects longitudinally goes beyond the scope of the present analysis. The mechanisms underlying the sustained excess mortality therefore remain uncertain. Frailty and multimorbidity are likely contributors,[22] but biological factors may also play a role. Notably, imaging studies have shown that brain atrophy rates after cSDH can exceed even those observed in Alzheimer disease, suggesting potential neuroinflammatory or neurodegenerative cascades that may accelerate cerebral aging.[23] At the same time, significant cerebral atrophy is often already present prior to the development of cSDH, especially in younger patients, suggesting that cSDH may instead reflect an advanced stage of preexisting neurodegeneration or frailty.[24] Descriptive analysis of causes of death within our cohort (eTable 1 in Supplement 1) revealed a balanced distribution: cardiovascular disease (26.8%), neurological causes (20.3%), and malignancy (18.3%) accounted for the largest proportions, a pattern that is broadly consistent with prior population-based studies of cSDH,[5,25,26] with no single cause of death dominating mortality. This reinforces the interpretation that cSDH represents a marker of increased biological vulnerability across multiple organ systems rather than an isolated condition with a uniform long-term trajectory.

Among long-term survivors, global QoL scores approached European population norms,[21] yet clinically relevant impairments persisted in multiple domains. CF and RF were significantly reduced in both sexes, while SF was significantly lower among men and showed a similar (albeit nonsignificant) trend among women. Importantly, all statistically significant impairments reached established minimally important difference thresholds for the EORTC QLQ-C30,[27] indicating that these deficits represent patient-perceptible impairments of clinical relevance beyond statistical significance alone. These findings point to higher-order limitations, particularly of memory, concentration, and social reintegration, which are unlikely to be captured by standard disability outcome scales. This multidimensional pattern aligns with previous observations of enduring impairments in attention, executive performance, and social reintegration after cSDH[6,12] and suggests that systematically assessing and addressing these higher-order outcomes should be a priority for future research and clinical care.

Taken together, the sustained excess mortality and cognitive and functional impairments observed in our cohort raise important questions about the broader significance of cSDH in late life. One possibility is that cSDH represents a sentinel health event,[3,4] or a turning point after which physiological and cognitive decline accelerate. In this view, the hematoma and its aftermath may initiate or amplify processes contributing to systemic and neurological deterioration. Alternatively, cSDH may serve as a surrogate marker of an already compromised health state, revealing preexisting frailty or multimorbidity that predispose patients to both reduced survival and diminished functional recovery. Either interpretation reinforces that cSDH marks a threshold of biological fragility that extends far beyond the perioperative phase.

From a clinical standpoint, our results emphasize the need for long-term continuity of care beyond hospital discharge with sustained engagement of primary physicians. Comprehensive postoperative follow-up beyond routine time points—including cognitive and frailty screening, neurorehabilitation or geriatric rehabilitation, and social reintegration programs—should be considered. Given that more than half of patients experienced preoperatively a recent fall with head trauma, and nearly three-quarters had some fall-related incident, incorporating mitigation strategies may be particularly relevant. This need is further underscored by the persistently reduced physical functioning reported by male survivors, indicating that mobility and daily functioning remain limited. Rehabilitation approaches similar to those established in stroke care involving multidisciplinary teams may serve as a useful model. From a health system perspective, this is particularly important given the marked increase in the incidence of cSDH over recent decades, which is expected to rise further with aging populations.[1,2] Establishing benchmarks for survival and HRQoL, as presented here, also provides a foundation for future cost-effectiveness analyses.

These findings also provide a pre-middle meningeal artery embolization (MMAE) benchmark against which future studies may compare long-term outcomes. As treatment strategies continue to evolve, including increasing use of MMAE, it will be important for future work to investigate whether such advances modify long-term survival or HRQoL outcomes.

### Strengths and Limitations

This study has several strengths. The precise mortality linkage through a national registry and the use of validated HRQoL instruments[19] ensure high data quality. In contrast to previous retrospective observations that included conservatively treated patients, our cohort consists of only surgically managed patients, ensuring a uniform treatment exposure and improving the interpretability of our findings. The control population[18] was rigorously matched by sex, age, and birth month, eliminating demographic bias. However, some limitations must be acknowledged. The study was conducted at a single center, which may limit external generalizability, and although patients were originally enrolled consecutively,[15] some degree of selection bias cannot be excluded. Regarding the survival analysis, it must be noted that the FSO dataset includes population movements and our cohort itself, but given the low migration rates in this mainly older cohort and offsetting effects of immigration or emigration, any resulting bias is expected to be minimal. As for the HRQoL analysis, approximately one-quarter of long-term survivors were lost to follow-up, introducing selection bias. Moreover, HRQoL was assessed cross-sectionally at 10 years rather than longitudinally, preventing evaluation of recovery trajectories over time. Finally, although normative comparisons were derived from a large, mainly European reference population,[21] this dataset did not include individuals from Switzerland, which may limit the cultural and demographic specificity.

## Conclusions

In conclusion, in this follow-up cohort study, patients who underwent surgical evacuation of cSDH experienced persistent excess mortality and self-reported long-term deficits in cognitive and functional domains a decade after treatment, despite largely preserved global QoL among survivors. The underlying mechanisms remain uncertain, highlighting the need for further research into the biological and systemic pathways that drive late mortality after cSDH. As endovascular and combined treatment strategies evolve, this study provides a critical benchmark for future comparisons and hints that improving long-term outcomes will likely require more than procedural innovation alone. Integrating neurorehabilitation, frailty management, and comprehensive postoperative care into treatment paradigms may be essential to extend survival and enhance quality of life in this growing patient population.

ARTICLE INFORMATION

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**Author Contributions:** Drs Petutschnigg and Goldberg had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Schucht and Goldberg contributed equally.
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*Acquisition, analysis, or interpretation of data:* Petutschnigg, Aschwanden, Descombes, Nasiri, Bervini, Murek, Häni, Jesse, Schär, Zubak, Z'Graggen, Schucht, Goldberg.
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*Supervision:* Murek, Zubak, Z'Graggen, Raabe, Schucht, Goldberg.

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**Data Sharing Statement:** See Supplement 2.

**Additional Information:** ChatGPT (OpenAI) was used between October 22, 2025, and February 12, 2026, to assist with language editing and improvement of clarity during manuscript preparation. Drs Petutschnigg and Goldberg take responsibility for the integrity of the content generated.

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